

Review

An Overview of Technical Aspects and Challenges in Designing Edge-Cloud Systems

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Abstract

Edge–cloud computing has emerged as a key enabling paradigm for augmented and virtual reality (AR/VR) systems because of the stringent computational and ultra-low-latency requirements of AR/VR workloads. Designing efficient edge–cloud systems for such workloads involves multiple technical aspects, including communication technologies, service placement, task offloading and caching, service migration, and security and privacy. This paper provides a structured and technical analysis of these aspects from an AR/VR perspective. We adopt a two-stage literature analysis, in which Google Scholar is used to identify fundamental technical aspects and solution approaches, followed by a focused analysis of recent research trends and future directions using academic databases (e.g., IEEE Xplore, ACM Digital Library, and ScienceDirect). We present an organized classification of the core technical aspects and investigate existing solution approaches, including heuristic, metaheuristic, learning-based, and hybrid strategies. Rather than introducing application-specific designs, the analysis focuses on workload-driven challenges and trade-offs that arise in AR/VR systems. Based on this classification, we analyze recent research trends, identify underexplored technical areas, and highlight key research gaps that hinder the efficient deployment of AR/VR services over edge–cloud infrastructures. The findings of this study provide practical insights for researchers and system designers and help guide future research toward more responsive, scalable, and reliable edge–cloud AR/VR systems.

Keywords: edge–cloud computing; network communication; service placement; offloading; caching; service migration



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1. Introduction

Integrating edge and cloud computing has significantly improved the resource-intensive systems because they allow complex computations close to the users, enriching their experience. Edge computing involves processing data closer to the source of data generation, typically at the network's edge, rather than relying solely on a centralized cloud server. Although edge computing handles tasks that require low latency, cloud computing can be used for resource-intensive operations [1–3].

Several technical aspects must be considered when designing the systems enabled by edge and cloud computing infrastructures. These aspects span a range of domains, including network architecture, communication technologies (wired or wireless), service placement solutions (offline, online, agile, etc.), task offloading/caching approaches (binary,

partial, etc.), service migration techniques (live or non-live), and security and privacy aspects [4]. A deep understanding of these technologies and technical considerations is essential to develop efficient edge–cloud systems.

For computationally intensive systems such as augmented and virtual reality (AR/VR), optimizing the use of edge–cloud infrastructures is crucial to meeting increasingly strict requirements related to scalable resource provisioning, security, and interoperability. Failing to effectively design the edge–cloud infrastructure for such systems can lead to significant latency spikes, inefficient resource utilization, and even service disruptions. In fact, to design an optimal edge–cloud infrastructure for computationally intensive systems, technical considerations must be systematically addressed to ensure high-quality service delivery. However, the existing research often focuses on isolated technical aspects, such as task offloading or service placement, without providing a comprehensive classification that encapsulates the broader design principles of edge–cloud systems. This paper aims to bridge this gap by presenting a structured overview of the key technical aspects involved in designing such systems to identify research challenges and propose future directions to enhance their efficiency and scalability. The motivation behind this work stems from the crucial need to guide system designers, researchers, and industry stakeholders in addressing the multifaceted technical challenges of edge–cloud deployment. By systematically categorizing key technical aspects (such as service placement, task offloading, caching, migration, and security/privacy), and proposing potential solutions, this study provides valuable insights to support informed decision-making. Through this paper, we address the following research questions (RQ):

- RQ1: What key technical aspects are described in the literature for designing edge–cloud computing systems with an emphasis on AR/VR applications?
- RQ2: What kind of techniques and solutions are studied in the literature for these technical aspects?
- RQ3: What are the key gaps identified in the existing research, and what future research directions are suggested for edge–cloud systems supporting AR/VR applications?

Answering RQ1 helps researchers identify each technical element in detail and better understand them before designing edge–cloud systems. It also allows researchers to identify the most significant effect of technical aspects and focus on optimizing critical components. Answering RQ2 helps researchers identify various types of techniques and possible solutions related to each technical aspect in a classified structure. Answering RQ3 empowers researchers to align their work with the evolving needs of the domain.

Despite the extensive body of work on edge and cloud computing, existing studies predominantly investigate technical aspects under general-purpose assumptions. However, AR/VR systems impose fundamentally different constraints, including ultra-low latency, continuous user interaction, and highly dynamic computational demands. As a result, solution strategies derived from general edge computing research cannot be directly applied to AR/VR-enabled systems without careful re-examination. The urgency of this problem is further amplified by the rapid deployment of AR/VR systems in industrial, educational, and collaborative environments, where delayed responses or service disruptions can severely degrade user experience or system reliability. Without a clear understanding of how core technical aspects must be adapted to meet AR/VR requirements, the design of responsive and scalable edge–cloud systems remains fragmented.

Unlike existing surveys that focus on general edge–cloud systems, this study specifically considers the unique challenges of AR/VR systems in edge–cloud environments. Although the prior studies mostly investigate individual components separately, this paper takes an integrated perspective and presents a systematic classification encompassing five key technical aspects of these systems. Furthermore, by reviewing the recent literature,

we identify ongoing challenges and current research trends, providing insights into areas where researchers and industry practitioners can direct their efforts to bridge existing technical gaps. In addition, we discuss potential solutions for efficiently managing these technical aspects, including heuristic, metaheuristic, and machine learning (ML)-based approaches. Through this structured study, this paper offers a clearer understanding of prevailing trends and future directions, distinguishing itself from previous reviews by providing a more comprehensive and application-specific perspective. It is worth noting that although the identified technical aspects in this paper overlap with those commonly studied in general edge computing, this work distinguishes itself by analyzing and structuring them explicitly according to the stringent real-time, computational, and interaction requirements imposed by AR/VR systems. Therefore, the main contributions of this paper are as follows:

- We provide a structured classification of core technical aspects in edge–cloud systems and investigate them under AR/VR-driven computational and latency constraints.
- We investigate solution strategies and trade-offs (heuristic, metaheuristic, learning-based, and hybrid) from a system-level perspective, highlighting their suitability for real-time AR/VR workloads.
- We analyze recent research trends to identify maturity levels and underexplored technical challenges, providing guidance for future research directions in AR/VR-enabled edge–cloud systems.

The rest of this paper is developed as follows. Sections 2 and 3 present related work and the research methodology of the article, respectively. Section 4 introduces the architectural paradigms for AR/VR applications, discussing their pros and cons. Section 5 categorizes the technical aspects of edge–cloud computing for AR/VR and explores various types of techniques. Section 6 discusses possible solutions for the technical aspects and Section 7 provides a literature analysis. Section 8 presents future research directions, followed by Section 9 that concludes the article.

2. Related Work

Based on our literature review, we identified two types of survey papers in this domain: (1) general survey/overview papers that broadly discuss edge–cloud systems without focusing on specific applications or implementations, and (2) AR/VR-specific survey/overview papers that focus on edge–cloud systems for specific applications.

2.1. AR/VR-Specific Surveys

Edge–cloud computing has been widely investigated as enabling paradigms for latency-sensitive and computationally intensive applications, particularly augmented and virtual reality systems. Several surveys have highlighted that the stringent delay and bandwidth requirements of mobile and immersive AR/VR workloads cannot be efficiently supported by cloud-only architectures. In this context, Siriwardhana et al. [4] provided a comprehensive overview of mobile augmented reality, emphasizing the critical role of 5G and edge computing in achieving ultra-low latency and high data rates. Their study categorized AR architectures into cloud-based, edge-based, localized, and hybrid solutions, illustrating how architectural choices directly influence responsiveness and user experience across diverse application domains. Building on this architectural perspective, other studies have examined the convergence of AR/VR with complementary technologies such as the Internet of Things and edge intelligence. Elawady et al. [5] explored the integration of IoT, edge computing, and AR/VR to support real-time interaction and latency reduction, highlighting the importance of distributed processing for immersive services. Similarly, Saini et al. [6] reviewed VR systems in IoT-enabled environments and discussed

enabling technologies such as object recognition, smart devices, and interconnected sensing infrastructures. These works demonstrate how edge-enabled architectures enhance the feasibility of AR/VR applications across domains including healthcare, education, military, and entertainment.

Beyond architectural design and integration, security and privacy have emerged as critical concerns in edge-assisted AR/VR systems. Ranaweera et al. [7] provided an in-depth review of security and privacy challenges in edge computing, identifying key threat vectors, vulnerabilities, and countermeasures associated with decentralized architectures; while not exclusively focused on AR/VR, their analysis highlights issues that become particularly critical in immersive applications due to continuous data collection, real-time interaction, and user mobility. Several studies have also further investigated AR/VR adoption in domain-specific and network-centric contexts. Eswaran et al. [8] examined AR/VR applications in manufacturing environments, identifying challenges related to hardware limitations, software integration, and system scalability. Lakshminarayanan et al. [9] studied the use of AR/VR and edge computing to improve healthcare accessibility in remote and underserved regions, emphasizing the role of distributed computation in reducing service latency. From a networking perspective, Dias et al. [10] investigated the integration of passive optical networks with edge computing to support latency-sensitive applications such as AR/VR, and explored the use of federated learning to improve resource allocation and reduce end-to-end delay. More recently, Loutfi et al. [11] analyzed mobility-aware computing in 6G networks, highlighting how mobile edge computing can better accommodate highly dynamic AR/VR workloads.

Existing studies provide valuable insights into architectures, enabling technologies, and application scenarios for AR/VR-enabled edge–cloud systems. However, most of these works either focus on specific application domains, individual technical aspects, or enabling network technologies, without offering a unified system-level analysis of the core technical challenges and their interdependencies. This observation motivates the need for a structured and comprehensive technical perspective, which is the focus of the present study.

2.2. Generic Surveys

A substantial body of literature has investigated task offloading, service placement, and resource management in edge, fog, and cloud computing environments, reflecting their central role in supporting latency-sensitive and computation-intensive applications. Several comprehensive surveys have focused on computation offloading as a core mechanism for reducing end-to-end latency and alleviating resource constraints at end devices. For instance, Lin et al. [12], Feng et al. [13], and Islam et al. [14] systematically categorized offloading strategies based on optimization objectives, task partitioning models, and decision-making mechanisms, highlighting challenges related to resource allocation, energy efficiency, and scheduling. More recent studies have extended this analysis by emphasizing intelligent and adaptive approaches, particularly those based on machine learning. Shakarami et al. [15], Taheri-abed et al. [16], and Rahmani et al. [17] examined learning-based and metaheuristic offloading strategies across cloud, fog, and edge environments, demonstrating their potential to improve decision quality under dynamic conditions.

Beyond task offloading, service placement and application deployment have been widely studied as complementary mechanisms that directly influence system responsiveness and scalability. Taleb et al. [18] and Salaht et al. [19] provided structured taxonomies of service placement algorithms in multi-layer edge–cloud systems, categorizing approaches based on optimization techniques, deployment models, and architectural assumptions. Related works have further explored AI-driven placement strategies, including evolution-

ary and learning-based methods, to address challenges such as resource heterogeneity, scheduling complexity, and energy efficiency [20–22]. In parallel, Malazi et al. [23] reviewed dynamic service placement approaches that jointly consider offloading, caching, and resource management, underscoring the need for coordinated decision-making across multiple technical aspects. Service migration and mobility management have also attracted growing attention, particularly in highly dynamic environments. Rejiba et al. [24] examined service migration strategies in edge computing, identifying migration scenarios, taxonomies, and architectural challenges related to user mobility and service continuity. Subsequent studies expanded this perspective by incorporating handover management and state migration in next-generation networks. Alkaabi et al. [25] and Jahandar et al. [26] investigated handover mechanisms in edge- and 6G-enabled systems, focusing on latency-aware migration and seamless application state transitions. These works highlight that mobility-induced dynamics significantly complicate placement and offloading decisions, especially for real-time services.

A related line of research has focused on resource scheduling and allocation, which underpins the effectiveness of offloading and placement strategies. Jamil et al. [27], Goudarzi et al. [28], and Rahimikhanghah et al. [29] analyzed scheduling mechanisms across cloud and edge layers, examining how task characteristics, environmental constraints, and optimization objectives affect system performance. Recent studies have further explored resource optimization in software-defined and AI-driven environments, particularly for real-time and IoT-oriented applications [30,31]. Complementary research on caching strategies has emphasized proactive and intelligent content placement to improve latency and bandwidth efficiency, as discussed by Aghazadeh et al. [32]. In addition to these core technical aspects, several surveys have examined broader architectural and system-level perspectives. Kumar et al. [33] analyzed edge–cloud architectures for Industrial IoT, while Carvalho et al. [34] and Rosendo et al. [35] provided comprehensive overviews of edge–cloud architectures and distributed intelligence frameworks. Other works explored domain-specific deployments, including smart agriculture [36], connected vehicles [37], UAV systems [38], and video streaming [39], demonstrating the versatility of edge–cloud computing across diverse application domains. Emerging studies have also investigated the convergence of edge computing with generative AI and advanced learning models to improve scalability and responsiveness [40–43].

Overall, existing studies provide extensive coverage of individual technical aspects such as task offloading, service placement, migration, caching, and resource scheduling. However, these works are often investigated in isolation or within domain-specific contexts, with limited emphasis on their interdependencies under stringent real-time and computational constraints. This fragmentation motivates the need for a unified, system-level analysis that connects architectural choices with core technical aspects, which is the focus of the present study. Table 1 shows a comparison of related work according to their focus and associated technical aspects.

Table 1. Comparison of related work according to their focus and relation to technical aspects.

Refs.	AR/VR-Specific	Offloading/Caching	Service Placement	Service Migration	Security/Privacy	Architectural Aspects
[4]	AR/VR-specific	✓	✓	✓	✓	
[5,9]	AR/VR-specific	✓				
[6]	AR/VR-specific		✓			
[7]	AR/VR-specific				✓	
[8]	AR/VR-specific					

Table 1. *Cont.*

Refs.	AR/VR-Specific	Offloading/Caching	Service Placement	Service Migration	Security/Privacy	Architectural Aspects
[10]	AR/VR-specific	✓	✓			
[11]	AR/VR-specific			✓		
[12–17,32]	Generic	✓				
[18–23]	Generic		✓			
[24–26]	Generic			✓		
[27–31]	Generic	✓		✓		
[33–37,39]	Generic					✓

2.3. Limitations of Existing Surveys

The majority of current overview/survey papers examine the technical aspects from a general perspective, with a limited number of articles focusing specifically on AR/VR-based use cases. Although the existing surveys provide valuable insights into edge computing and AR/VR-enabled systems, they mostly fall into two categories. The first category focuses on general edge or cloud computing frameworks, where technical aspects are studied under generic assumptions. These studies do not explicitly account for the stringent computational and ultra-low-latency requirements imposed by AR/VR workloads. The second category concentrates on specific AR/VR applications or individual technical aspects, such as task offloading or communication technologies, without providing a unified system-level perspective. As a result, the interdependencies and trade-offs among multiple technical aspects remain insufficiently explored.

Many surveys, such as those by Siriwardhana et al. [4] and Elawady et al. [5], discuss the role of AR/VR in conjunction with enabling technologies like edge computing, IoT, and 5G. However, these works mainly emphasize high-level architectural considerations and application domains rather than technical methodologies for resource allocation, computation offloading, or service placement in these environments. Moreover, the surveys that focus on AR/VR-specific challenges often address only a limited subset of issues. For example, Eswaran et al. [8] discuss AR/VR in manufacturing but do not provide an in-depth analysis of technical and architectural aspects for AR/VR workloads. Similarly, Lakshminarayanan et al. [9] examine AR/VR applications in healthcare but do not extensively analyze performance trade-offs in edge–cloud orchestration. In contrast, most generic surveys in edge computing, such as those by Lin et al. [12] and Islam et al. [14], present comprehensive analyses of resource management, scheduling, and service placement but do not explicitly consider the stringent requirements of the specific applications. These surveys tend to generalize edge computing challenges without explicitly addressing the unique demands of specific applications.

This study addresses these limitations by providing a structured, workload-driven analysis of core technical aspects in edge–cloud systems, explicitly framed around AR/VR-driven constraints. Rather than proposing new models or algorithms, the contribution lies in restructuring and critically analyzing existing solution spaces to identify applicability boundaries, research gaps, and future directions specific to AR/VR-enabled edge–cloud systems.

3. Research Methodology

Given the broad scope of edge–cloud systems and the numerous studies published over the past years in this domain, we employ two distinct search methodologies to identify relevant papers and works for our study in this paper. The first methodology highlights several significant works related to architectural models of AR/VR systems and technical aspects of edge–cloud AR/VR systems. The second methodology explores recent research trends related to each architectural and technical aspect. We did not attempt to cover all the

papers comprehensively for our first search methodology because our aim is to determine the significant works that are more closely related to the domain.

In line with our aims, we focused mainly on relevant works related to technical aspects and their associated techniques. To achieve this, we conducted searches using Google Scholar. We initially set the date range to the last decade and defined specific keywords for each aspect. After reviewing the papers, we prioritized journal papers over conference papers and selected the most relevant and recent ones to study in this paper. Given that Google Scholar prioritizes more relevant papers, we investigate the first 100 papers in more detail, including those cited by these papers. This example was also applied to other technical and architectural aspects. Sections 4–6 are based on the first search methodology.

Due to the extensive body of work resulting from the first search methodology (which made it impossible to investigate all papers in detail), we focused on a more limited but prominent selection of libraries in the second search methodology to review recent contributions and trends in the domain. We considered three primary reference sources: IEEE, ACM, and ScienceDirect. Our search for related references included journal and conference papers published during the last five years. We focus on the last five years publications to rigorously assess recent trends in the field. For the second search methodology, we also used a specific search string for each aspect. For example, the search string and keywords used for service placement was “(augmented reality OR virtual reality) AND (edge computing OR fog computing OR edge-cloud computing) AND (service placement)”. For other technical aspects, we utilized a similar search string and applied the query to the subject and abstract of the articles.

The initial output of this stage yielded several hundred papers across all aspects. These results were then filtered using the following inclusion criteria: (i) peer-reviewed journal or conference papers, (ii) explicit focus on technical or system-level aspects of edge–cloud computing, and (iii) relevance to latency-sensitive or computation-intensive workloads (including AR/VR). Papers were also excluded if they: (i) focused solely on application demonstrations without technical analysis, (ii) lacked relevance to edge–cloud architectures, or (iii) duplicated content already covered by more comprehensive studies. After applying these criteria and removing duplicates across queries and databases, the final dataset formed the basis for the trend analysis presented in Section 7, where 249 unique papers remained. Subsequently, 179 papers were selected for detailed analysis after applying the inclusion criteria, of which 73 papers specifically addressed AR/VR-related technical aspects, while 106 papers focused on generic edge–cloud computing aspects. A total of 70 papers were excluded due to insufficient technical or system-level focus, and additional papers were excluded because they primarily presented application demonstrations without providing substantive architectural or methodological analysis.

4. Architectures for AR/VR Applications

AR/VR architectures can be classified into cloud-based AR/VR, edge-based AR/VR, and edge–cloud AR/VR. The classification is based on the specific location where AR/VR processing and services are run.

4.1. Cloud-Based AR/VR Architecture

In a cloud-based AR/VR system, the architecture is modeled as server–client, where the end devices (clients) are responsible for running the final application within the system, and the AR/VR server is located in the cloud so it can be easily accessed from any location. In a cloud-based AR/VR, after collecting data and capturing videos or images from the environment, the end devices perform some mandatory pre-processing at the device level and then send the captured data to the AR/VR server. In the cloud, the most complicated

computational tasks are executed by the AR/VR server. A cloud-based AR/VR architecture is depicted in Figure 1a.

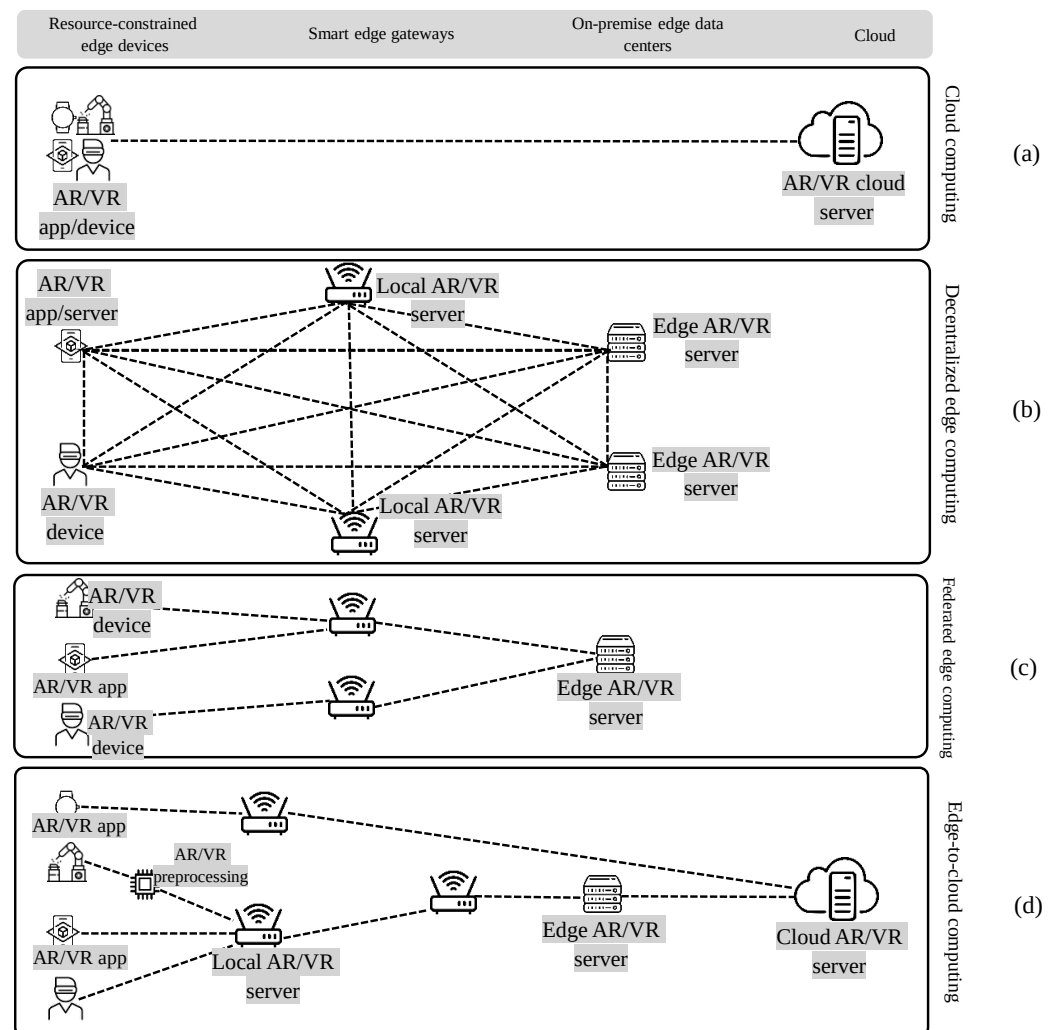


Figure 1. (a) Simple cloud computing architecture. (b) Decentralized edge computing architecture. (c) Federated edge computing architecture. (d) Edge–cloud computing architecture.

In cloud-based AR/VR systems, the server has substantial computational resources to manage the high processing demands. Offloading intensive computing AR/VR tasks (e.g., rendering complex 3D graphics) to the cloud enables resource-constrained devices to focus on lighter tasks (e.g., handling user input, managing communication with cloud servers, and displaying the results). In a cloud-based AR/VR system, upgrading the fundamental image/video processing software would be easier when executed in a central location because users or system administrators can update the software in one central location, rather than update it on individual devices. Furthermore, handling failures or faults related to the core of AR/VR-based software in a central place is simpler. Despite these advantages, communication delays remain a major challenge in cloud-based AR/VR architecture due to the substantial volume of data transmitted from the end AR/VR devices to the cloud. Lower reliability and degree of security and privacy are other disadvantages of cloud-based architecture, considering that sensitive data could sometimes be transmitted to remote places. Cloud-based systems may face downtime or connectivity issues as well, affecting the reliability of the system as a whole; mainly because the system relies on remote servers and network connections that may not always be available or stable. Consequently, cloud-based architecture is not suitable for AR/VR applications that require real-time

monitoring of systems and sensitive data [44]. These limitations make purely cloud-based architectures unsuitable for latency-critical and interactive workloads, motivating the need for computation closer to the user.

In the literature, several works focused on cloud-based AR/VR for various applications. For example, Dahbi et al. [45] designed an open-source cloud-based AR/VR application called “CollaBIM” that facilitates visualization, collaboration, and interaction capabilities within the context of a Building Information Modeling (BIM) model. Through its cloud integration, the application operates as a connection point that allows projects and team members to obtain real-time information about project developments, view models, and interact with project models/objects. Roche et al. [46] introduced a cloud-based AR model for shopping systems. The introduced system contains two primary components: a Web App client and a cloud server. The Web app allows users to capture images/videos and send them to the cloud server for rendering and/or running analytical operation on the captured data. Khan et al. [47] introduced a cloud-based mobile AR for IoT use cases. They claimed that their AR platform can be used for different IoT scenarios so that mobile devices (smartphones, watches, etc.) can capture images/videos and transfer them to a cloud server for further processing according to the objectives of the IoT use case. Alkevi et al. [48] also introduced a cloud-based VR application for healthcare applications. The work discussed Virtual Reality Exposure Therapy (VRET) and investigated whether placing patients with mental problems in a simulated virtual 3D environment can significantly affect their treatment. To this end, they modeled a cloud-based VR architecture to create a lightweight mobile solution that any psychology clinic can efficiently adapt. Abichandani et al. [49] developed an application for cloud-based VR systems for education. This work designed a cloud-based VR to provide students with a comprehensive and immersive learning experience in the domain of solar energy and photovoltaic technology. Franzluebbbers et al. [50] worked on a cloud-based VR platform for laser scans of plants.

While several cloud-based AR/VR research projects across various applications have been introduced in the literature (some presented in this article), recent studies in this area are shifting focus away from cloud-based solutions due to concerns related to high transmission delay, security, and privacy.

4.2. Edge-Based AR/VR Architecture

Edge computing can be implemented in either a decentralized or a federated manner. In decentralized edge-based AR/VR, all devices, including resource-constrained devices, edge gateways, and edge on-premise servers, are connected in a peer-to-peer manner. In such a model, resource-constrained devices can run AR/VR applications and perform image processing computations. In addition, the devices can connect to other AR/VR servers in the edge network for further processing. In this architecture, each server offers distinct AR/VR services and each client connects to a particular server to access the specific service that meets its requirements with respect to peer-to-peer communications. Figure 1b shows the decentralized edge-based AR/VR architecture and indicates how different entities connect and work together. Federated edge-based AR/VR platforms work in a master–slave or client–server manner as well, similar to the model used in cloud computing. Figure 1c shows an overview of federated edge-based AR/VR, where the server(s) are generally arranged in the edge layer near the end devices.

One of the primary advantages of fully decentralized models is their robust hardware reliability. These models are not susceptible to interruption or failure, even when parts of the system encounter faults and errors in processing AR/VR tasks [51]. In terms of privacy and security aspects, data remain confined to the local area in the fully decentralized architecture, and because data are not transmitted to remote central locations in this model,

there is a lower risk of unauthorized access or interception during data transmission. End users also have more control over their data, as it remains within their own environment. Therefore, the decentralized model reduces its vulnerability to security concerns and minimizes privacy risks [52]. In terms of communication aspects, most previous work prefers to use federated edge computing because fully decentralized models often incur higher communication costs (as there is sometimes a need to transmit and synchronize multimedia content across various nodes). From an energy consumption perspective, battery consumption within the decentralized edge-based model is also expected to be high, particularly due to processing tasks involving images and videos that may need to be executed on battery-constrained devices. A disadvantage of both models is that complex AR/VR applications running on resource-constrained edge devices or edge servers can lead to increased energy consumption and execution time of tasks. For example, computational tasks such as rendering high-definition graphics, tracking user movements, and processing multimedia data in real-time are computationally intensive and require significant computational power and memory. When these tasks executed on resource and energy-constrained edge devices or servers, it leads to increased energy consumption and longer execution times of the applications. However, edge-based systems significantly reduce transmission time because they do not transfer data to remote servers. Another drawback, especially in a fully decentralized model, is that duties related to updating and upgrading the software and addressing technical issues require individual attention and action on every device [52]. Overall, although edge-based architectures mitigate latency, their limited computational capacity and management complexity pose challenges for large-scale or computation-intensive AR/VR services.

Several studies in the literature are dedicated to decentralized edge-based AR/VR and federated edge-based AR/VR systems. However, most of the research on decentralized edge architecture focuses on scenarios where security and privacy are important within the system, while other metrics, such as latency, are considered as secondary priorities [52]. The most applicable use cases for a decentralized model emerge when AR/VR is integrated with blockchain technology [53–56]. The main objective of integrating AR/VR and blockchain is to handle secure communication protocols and aid in virtualization, privacy, and enhancement of remote computational resources. Zhou et al. [57] introduced a federated edge-based system for AR-based vehicle ad hoc networks, where base stations near roads are considered computational units. Similarly, Akter et al. [58] utilized federated edge-based architecture for AR/VR applications for assisting blind and visually impaired people with the aim to improve the battery consumption of the AR/VR devices. Bartolomeo et al. [59] conducted extensive experiments on a distributed stream processing-based AR pipeline using an edge-based architecture. They provided useful and practical insights for leveraging edge computing in AR applications, specifically for large-scale AR operations. Kassinger et al. [60] designed an AR-based simulation environment for medical applications based on edge-based architecture to enable real-time simulations. In another work, Kassinger et al. [61] utilized an edge-based AR architecture to model and simulate biophysical-based problems, where the computations for these simulations were performed on edge devices or servers. Mulloni et al. [62] presented a federated edge-based model of an AR-based interface in indoor navigation applications [63], with a local user to perform calculations related to the AR application of users inside the building. Chuang et al. [64] also investigated the latest advances in edge-based caching and computing in AR/VR applications, focusing on mobile AR/VR applications. Despite several research efforts on edge-based AR/VR, this architecture becomes impractical when AR/VR applications involve a large amount of computations and/or data.

4.3. Edge–Cloud AR/VR Architecture

An edge–cloud AR/VR architecture defines a cutting-edge framework to deliver immersive AR/VR experiences to end users. This architecture is used in most recent works, combining cloud and edge capabilities to optimize latency, reliability, scalability, and energy consumption, as well as to enhance security and privacy for AR/VR applications. The edge–cloud model for AR/VR applications is depicted in Figure 1d, where the cloud server is a central place for processing massive amounts of AR/VR data and providing powerful computational resources. In addition, edge components can act as local servers when the applications require low latency or critical updates quickly.

The edge–cloud architecture inherits the advantages of edge and cloud computing. This integration improves its ability to reduce end-to-end latency, making it a more relevant architecture in the literature. This model reduces data transmission across the network and minimizes critical performance indicators in AR/VR applications, such as those related or sensitive to latency and reliability. From a security and privacy viewpoint, the edge–cloud model provides a balance between the benefits of edge-based and centralized cloud-based models, but may not deliver the same level of security and privacy as pure edge-based models (especially fully decentralized edge-based architecture) provide. The edge–cloud model reduces the security and privacy concerns by reducing data transmission to remote places; however, if the cloud is assumed to be the central coordinator of the entire edge–cloud network, the system becomes vulnerable to security attacks and failures. Moreover, the flexibility and scalability of this model allow end devices to access additional computational and storage resources from edge or cloud servers. This architecture also improves energy consumption compared to decentralized edge models due to federated communications. However, one of the main drawbacks of this model is its complexity as compared with cloud-based and federated edge-based models. This complexity arises from the existence of multiple AR/VR servers in both the edge and cloud layers, leading to challenges in managing, controlling, and synchronizing the network for specific applications. [65,66]. Overall, by balancing latency reduction with computational scalability, edge–cloud architectures provide a more flexible foundation for supporting demanding AR/VR workloads.

In recent work, edge–cloud architecture has become the preferred choice for deploying AR/VR applications. For instance, Didar et al. [67] introduced an AR framework that uses the edge of the network in combination with cloud computing to improve the energy consumption of AR applications. The main objective of this work is to optimize energy consumption while simultaneously reducing the load on AR devices. Shi et al. [68] introduced and executed a concept, called Mobile Edge Computing-VR (MECVR), to employ an edge–cloud architecture for running and delivering VR services on low latency mobile devices. Liu et al. [69] proposed an edge–cloud architecture for object detection in AR applications. In their work, the main aim was to improve the latency and bandwidth usage and improve energy consumption. They designed an edge–cloud AR/VR system that allows high-accuracy object detection for commodity AR/VR systems, while using the ability of the edge devices to detect complex objects in the environment with minimum latency. Chen et al. [70] utilized an edge–cloud infrastructure for AR applications, focusing on addressing latency concerns in the object detection process of AR applications. Similarly, Zhang et al. [71] investigated video stream generation for multi-party mobile AR applications using an edge–cloud infrastructure, aiming to maximize the user experience quality and minimize service costs. Younis et al. [72] implemented a demo for mobile AR applications, using an edge–cloud infrastructure. They used smartphones and embedded devices (e.g., Raspberry Pi) as edge devices to run some computations related to AR applications on these devices. There are also other recent works for designing edge–cloud AR/VR systems, while considering bandwidth usage and latency of edge–

cloud AR/VR (e.g., [68,73–77]), as well as others focusing on both energy consumption and latency/bandwidth of edge–cloud AR/VR systems (e.g., [78–80]).

The above architectural analysis highlights that no single deployment model can fully satisfy the computational and latency requirements of AR/VR systems in isolation. These architectural trade-offs directly influence higher-level technical decisions, such as where services should be placed, which tasks should be offloaded, and how migration should be handled. Consequently, the following section focuses on the core technical aspects that arise from these architectural choices.

5. Technical Aspects of Edge–Cloud AR/VR Systems

This section provides an organized classification of the technical aspects involved in designing edge–cloud AR/VR systems. Based on our literature review, we identify five key categories: communications, service placement, task offloading/caching, service migration, and security/privacy aspects. Each category is subsequently discussed, highlighting the challenges it poses to the design of AR/VR systems. We provide an in-depth analysis of each category and propose various types of techniques (subcategories) that can be used for each technical aspect. The classification proposed for these technical aspects is illustrated in Figure 2 and discussed in the following subsections. Although the high-level technical aspects align with those studied in general edge computing, their classification and analysis in this paper are tailored to AR/VR systems, where latency sensitivity, workload dynamics, and real-time interaction fundamentally reshape design trade-offs. The technical aspects discussed in this paper are common across all AR/VR applications. They are fundamentally driven by shared workload characteristics, including computational intensity, strict latency constraints, and dynamic execution environments, rather than by differences in application domains.

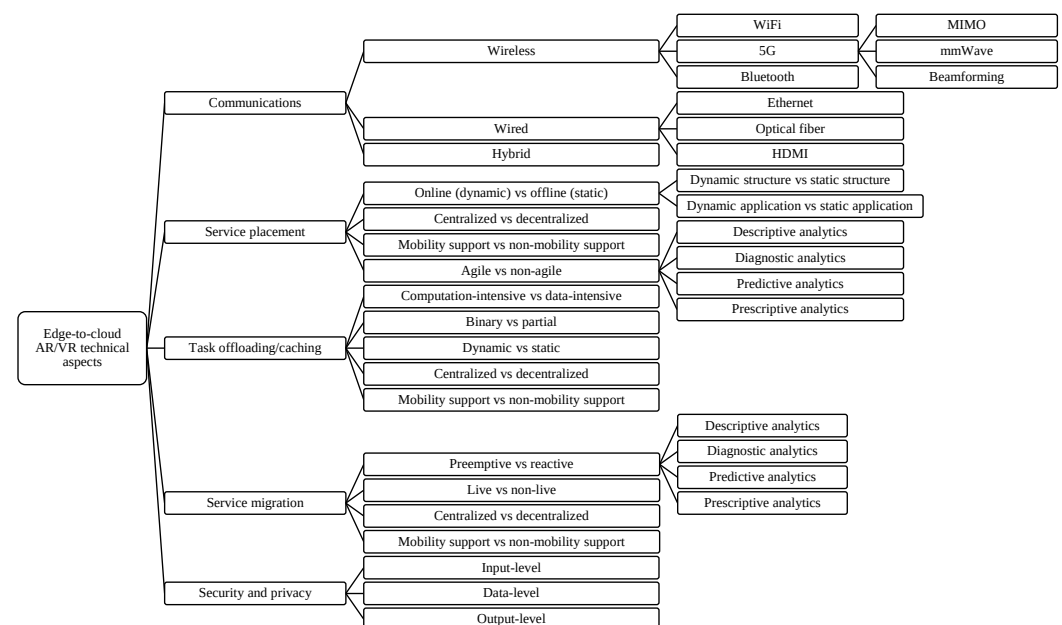


Figure 2. Technical principles in edge–cloud AR/VR.

5.1. Communication Technologies for Edge–Cloud AR/VR Applications

In AR/VR, network communications are especially critical for downloading or streaming high-quality content and supporting multiple interactions [81]. Because AR/VR applications often require continuous updates, upgrades, and access to cloud-based resources, these tasks depend on low-latency, reliable, stable, and energy-efficient communications [82]. The complexity arises when there are many available communication

technologies (wired, wireless, or both), each with unique attributes and suitability to different contexts. Consequently, picking the most appropriate communication technology to optimize AR/VR performance and meet the overall QoS of the system is problematic. However, this choice is not exclusively a choice of one specific technology, and different communication technologies can be selected for different parts of the network. For example, communications between edge devices and edge servers can be done using wireless communications, and communications between edge servers and the cloud can be done using wired communications. Therefore, understanding how these communication technologies interact with the systems and how they impact the user's experience is complicated on both the edge and the cloud sides. Furthermore, the dynamic aspect of communication technologies and the growing expectations of users over time become another problem that should be considered. Communication requirements and constraints grow as AR/VR applications expand across various domains. However, communication technologies have also been advancing over time to meet the expectations of users. For example, 4G marked a significant leap forward with faster data speeds and lower latency. After 2019, 5G technology appeared with even faster data speeds, extremely low latency, and the ability to connect many devices simultaneously. Now, 6G is expected to become a more robust wireless communication technology than previous generations [83].

Therefore, designing a comprehensive platform that combines decision making, real-time optimization strategies, and user-centric feedback is important to continuously refine connectivity and communication choices and ensure that AR/VR technologies reach their full potential. Various technologies are used for communications in current edge-cloud AR/VR applications, as discussed below.

5.1.1. Wireless Communications

Various kinds of wireless communication are used for AR/VR. Wi-Fi, represented by the 802.11 standards, ref. [84] provides a smooth and reliable connection within specified spaces, making it the most suitable option for streaming video to locally connected servers. Indoor AR/VR applications are examples of this wireless communication technology that is commonly used because of its reliable data rates and moderate latency [85].

5G is also assumed to be a preferred wireless technology for AR/VR (due to its high-speed and ultra-low-latency characteristics). 5G and emerging 6G systems introduce system-level features that directly influence the feasibility and efficiency of AR/VR deployments. Among these, Ultra-Reliable Low-Latency Communication (URLLC) plays a central role by enabling bounded latency and highly reliable packet delivery, which are essential for maintaining immersion and preventing user discomfort. Another key enabler is network slicing, which allows multiple virtual networks with distinct quality-of-service guarantees to coexist on a shared physical infrastructure. For AR/VR applications, network slicing enables the isolation of immersive traffic from best-effort services, ensuring predictable latency, bandwidth, and reliability. Looking beyond 5G, 6G networks are expected to further strengthen support for AR/VR workloads by providing tighter latency bounds, integrated sensing and communication, and native AI-driven network management. These advancements are anticipated to enable more adaptive and autonomous orchestration mechanisms to facilitate real-time service placement, task offloading, and migration under highly dynamic conditions.

Other low-powered options, such as Bluetooth, are generally deemed inappropriate for AR/VR applications. For example, Bluetooth (as a low-power wireless technology) is commonly used for short-range connections. Although Bluetooth technology is useful for low-battery devices and facilitates seamless integration of controllers, headphones, and other devices with AR/VR hardware, it is not the preferable choice for connecting

AR/VR devices with edge devices and servers because of its limited range (typically up to 100 meters depending on the Bluetooth class) [86]. It also introduces higher latency compared to other wireless technologies.

Besides various benefits (e.g., outstanding mobility and high scalability) that make wireless communications readily adaptable for the continuous growth of AR/VR devices/applications, they could also expose a serious burden to the system due to their inherent communication latency. To that end, new technologies are emerging to overcome the latency problem, making them suitable options [87,88]. For example, Massive Multiple-Input Multiple-Output (MIMO) technology in 5G utilizes many antennas on both the server side and the end devices, to simultaneously transmit multiple data streams. Millimeter Wave (mmWave) is also a communication technology that operates in the frequency range between 24 and 100 GHz and offers quite high data transfer rates, which are essential for AR/VR applications. Another technology is beamforming, which focuses on transmitting and receiving signals in a specific direction rather than broadcasting signals in all directions. When these new technologies are integrated into a 5G network, they collectively provide high throughput, low latency, scalable, and reliable connections for AR/VR applications [4].

5.1.2. Wired Communications

Different wired communication technologies are used for AR/VR. Ethernet is one of the most prevalent choices, well known for its reliability and stability, as well as its ability to deliver consistent high-speed data transmission. Different Ethernet technologies can provide network speeds ranging from 100 Mbps to 100 Gbps. Fiber optics appears to be another technology for wired connectivity. This technology is famous for its unparalleled data transfer and latency capabilities. This technology transfers data modulated as light signals, resulting in highly fast speeds and extremely low latency. The High-Definition Multimedia Interface (HDMI) is another wired technology that is used to connect AR/VR devices, to headsets, computers, displays, or gaming consoles. This technology is well known for delivering high-quality audio and video signals that can be chosen for communication at the device level in indoor AR/VR environments [89].

Wired technologies are used for their low-latency communications. They provide high data rates with reliability when compared with wireless communications. However, wired connections require additional infrastructure (cables, connectors, etc.) for setting up AR/VR systems, and thus are neither swift nor cost efficient for deploying AR/VR systems.

5.1.3. Hybrid Communication

Due to specific pros and cons of both wired and wireless technologies, a hybrid connection policy seems to be a reasonable and preferred solution for edge–cloud AR/VR systems. The choice of hybrid network connections depends on the limitations and purpose of the network. For example, if energy consumption is an important factor in the AR/VR network, the arrangement of hybrid communications would differ from when latency is the first priority. Therefore, it is important to understand all performance metrics before prioritizing them and eventually striking a balance between them. Figure 3 shows an example of general hybrid communications for edge–cloud AR/VR systems, where Bluetooth is used for device-level connections, Wi-Fi for connecting AR/VR devices to edge servers, and Ethernet/5G/Fiber for connecting edge servers to cloud servers. In most research studies, network communication policy is not typically considered a primary objective for AR/VR systems. However, numerous works utilize wireless communication, especially 5G, in their network platforms for their AR/VR applications (e.g., [82,90–96]). A few studies use wired technologies for communication (e.g., [6,63,97]).

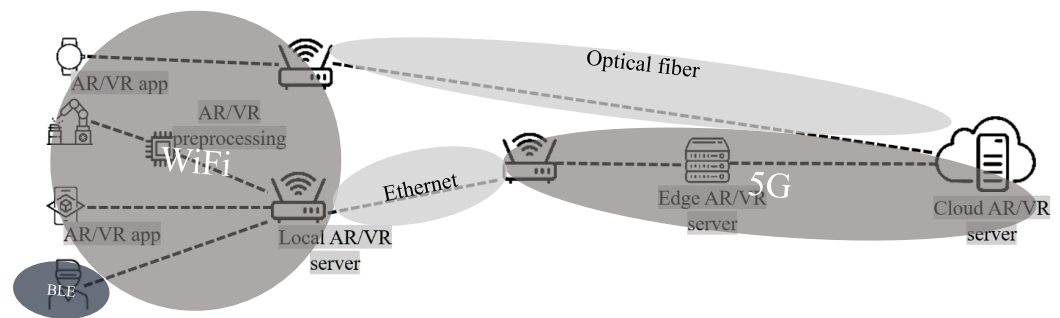


Figure 3. A typical hybrid communications network for edge–cloud AR/VR systems.

Although end-to-end latency is commonly used as a primary performance indicator in AR/VR-enabled edge–cloud systems, it implicitly reflects a broader set of QoS requirements, including delay jitter, packet error rate (PER), throughput, and real-time or near real-time responsiveness. In practice, these QoS dimensions are tightly coupled and highly dependent on deployment scenarios, network configurations, and traffic characteristics. Therefore, this survey discusses wired and wireless technologies at an architectural and system-design level, focusing on their ability to support latency-sensitive and reliability-critical AR/VR workloads rather than prescribing technology-specific QoS thresholds. It is also worth noting that low-power wide-area and short-range IoT technologies (e.g., IEEE 802.15.4 [98], ZigBee, and LoRa) are primarily designed for low-data-rate, delay-tolerant services and therefore do not satisfy the bandwidth, latency, and reliability requirements of AR/VR workloads.

5.2. Service Placement

Service placement is another technical aspect when designing edge–cloud AR/VR systems. It involves determining the optimal computing nodes for executing specific services to enhance system performance in terms of latency, reliability, energy consumption, throughput, scalability, etc. Given the complexity and heterogeneity of edge–cloud environments, optimal decision-making regarding how services should be placed on available resources across the edge–cloud network can be very challenging (in fact identified as an NP-Hard problem) [23,99–101].

To formulate service placement problems, interconnected graphs are preferred to indicate the service components of an application, where the vertices denote the processing and computational components, and the edges denote the inter-dependencies and communication requirements among these service components. Among several graph typologies (linear graph, a tree application graph, etc.) designed for this context, directed acyclic graphs (DAGs) are the most commonly employed choice due to their versatility in modeling real-world AR/VR applications. Figure 4 shows an example DAG that represents relationships between the service components of an AR/VR application. Figure 5 shows two different service placement policies in an edge–cloud network for the example DAG.

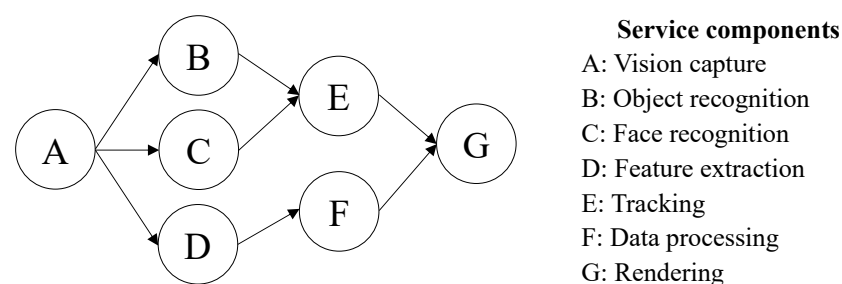


Figure 4. A simple example of a DAG of an AR/VR application.

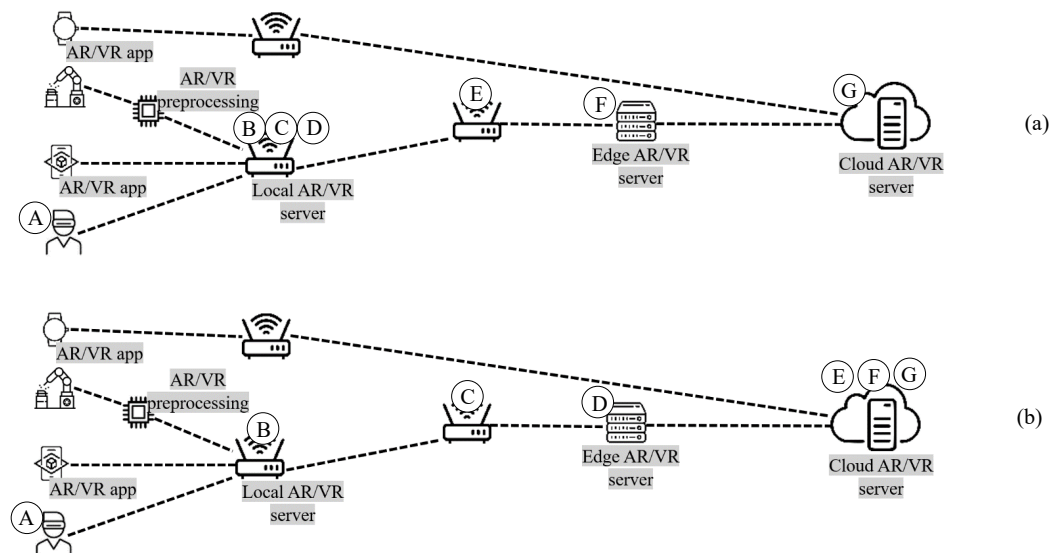


Figure 5. Sub-figures (a,b) show two different examples for service placement of an AR/VR application in an edge–cloud architecture.

Service placement techniques can be classified into different categories, according to the way they are supposed to work and function.

5.2.1. Offline vs. Online Service Placement

The service placement can be performed in two different modes: offline and online. Offline placement makes management and deployment decisions when complete information about a system (number of users, number on edge-nodes, capacity of link among nodes, etc.) is available. In contrast, online techniques [102,103] involve making decisions in real-time during system execution. These decisions are made based on both process characteristics (computational requirements, storage requirements, bandwidth needs, etc.) and the current condition of the system (available memory, network bandwidth, CPU, etc.) [104]. Researchers claim that many real-world systems are considered online problems, and thus algorithms must be able to handle online services. On the other hand, although online service management techniques can handle dynamic circumstances, they may not always utilize system resources efficiently (i.e., make optimal decisions). Therefore, the choice of using offline or online techniques greatly depends on the availability of information and the need for adaptability. Even though the online techniques can handle the system dynamics, they may not always optimize the system to the level the offline ones can.

In many research works, offline and online techniques are also called, respectively, static and dynamic techniques. Dynamicity in this context can be viewed from two aspects: dynamicity of the infrastructure and dynamicity of applications. Infrastructure dynamics are associated with the continuous change of compute/network entities (as they may arbitrarily join or leave the system) and fluctuations in the capabilities of available resources within the infrastructure over time. Application dynamics, on the other hand, arise from changes in real-world conditions of a system: for example, changes in service component dependencies, arrival or termination of access to new data sources, and access to real-time decisions by users to start or stop sending their data [103,105,106]. To capture all the complexities mentioned, the dynamic nature of both infrastructure and applications should be carefully considered when designing robust service placement techniques.

5.2.2. Centralized vs. Decentralized Service Placement

A centralized technique [100,107–112] controls the service placement process from one central entity or location, while a decentralized technique [113] distributes the control tasks

between multiple entities or locations. The centralized service placement techniques are used (more common) more than the decentralized ones, for the primary reason that it is much easier to find globally optimal solutions when using centralized techniques. They however cannot be easily scaled (low scalability) to solve very large problem instances. This is because as the size or the complexity of networks and/or services grow, centralized systems cannot cope with the computational complexity (e.g., convergence time) to process large amounts of information in reasonable timeframes. In contrast to centralized techniques, a decentralized technique involves multiple authorities and/or orchestrators to manage the network and make decisions collaboratively for service placement. This type of service placement better preserves privacy, as decentralized decision-making relies on local information. Because they are inherently more scalable, they can better adapt to the dynamic changes of edge–cloud environments. However, since each node makes decisions based on local information, overall service placement decisions may not lead to globally optimal (or near optimal) solutions [114].

5.2.3. Non-Mobility vs. Mobility Support

Service placement techniques that are run without considering changes in user mobility are called non-mobility techniques. Because these techniques are not inherently designed to handle mobility, they could face difficulties to control/handle increased latency caused by users when they move from the initial deployment location. A service placement technique that can manage frequent changes in the places of end devices (such as smartphones) is able to respond to various network-related problems (e.g., excessive delays or packet loss). Therefore, addressing end-device mobility is a key factor in service placement strategies for edge–cloud AR/VR systems when mobility is a crucial attribute of the end users [115].

5.2.4. Non-Agile vs. Agile Service Placement

In AR/VR-based use cases, the system must respond to users' actions, as well as to software and hardware changes in real-time to ensure that interactions with virtual objects or environments feel natural and immediate. Delays in response can break immersion and disrupt the system. Agile techniques mostly focus on developing intelligent algorithms (e.g., reinforcement learning-based algorithms) and parallel optimization methods (e.g., parallel metaheuristic algorithms) that respond quickly to abrupt changes. The primary objective of agile techniques [114,116–118] is to ensure the robustness and resilience of the system in response to unexpected situations. The data analytics used by agile techniques is categorized into four types [119]: *descriptive analytics*, *diagnostic analytics*, *predictive analytics*, and *prescriptive analytics*. Descriptive analytics involves interpreting historical data to gain insight into past performance and service-related events. A comprehensive understanding of services and system behavior can be obtained by analyzing historical data so that patterns, anomalies, and/or periodic issues can be identified. Diagnostic analysis seeks to answer why an issue occurs, and based on that, identify root causes of service disruptions. By understanding the causes, one can take action to address and prevent similar issues in the future. Predictive analytics uses historical data and statistical models to forecast future events. For example, an agile solution can use predictive analytics to anticipate service demand, resource requirements, and potential problems. Finally, prescriptive analytics improves predictive analytics by suggesting specific actions or strategies to optimize service management. Prescriptive analytics predicts what might happen first and then suggests the best possible responses for optimal outcomes [119].

5.3. Task Offloading and Task Caching

AR and VR technologies require significant computational power to deliver real-time experiences. If these computations are performed on resource-constrained edge devices, QoS expectations could be negatively affected. There are two ways to improve the QoS: task offloading and task caching. Through task offloading, computations from resource-constrained devices are migrated to more powerful edge devices, edge servers, or the cloud, aiming to enhance the overall performance and usability of AR/VR devices. Task caching, on the other hand, refers to methods where data are placed (or copied) closer to end devices. Task caching is also as important as offloading because storing frequently accessed data at edge nodes closer to the end user significantly reduces the latency of retrieving data from distant cloud servers [120].

Alongside numerous benefits of task offloading and data caching, there are some challenges when using them in AR/VR applications [120–122]. Latency issues are important challenges when network communication introduces delays. Bandwidth consumption and potential congestion, privacy, and security risks in transmitting sensitive data are other challenges that must be addressed. Moreover, designing optimal algorithms and policies is crucial to achieving task offloading/caching benefits. For example, consider an algorithm that offloads tasks without considering their priority based on a First Come First Out (FCFS) policy. In such a scenario, critical tasks may be delayed or queued behind less important ones, leading to performance issues. The same applies to caching. In this case, a latency- or deadline-aware algorithm can make optimal decisions about where and when to offload and cache tasks compared to an FCFS-based algorithm. Therefore, designing optimal algorithms for offloading and caching AR/VR tasks so that AR/VR-based applications can run effectively, with minimum delay and maximum reliability, is essential.

Similar to service placement, decision-making for offloading and caching can also be performed centralized or decentralized. In centralized approaches, offloading and caching decisions are managed by a central entity, typically in the cloud. In decentralized approaches, making decision about offloading and caching are distributed among multiple edge devices and nodes within the network. Offloading and caching techniques can also be designed to support the mobility of end devices and the dynamic nature of the entire system. Despite their many similarities with service placement techniques, they also have their own set of specific challenges such as computation-intensive vs. data-intensive offloading/caching and binary vs. partial offloading/caching techniques, which are discussed in the following.

5.3.1. Computation-Intensive vs. Data-Intensive Offloading/Caching

Computation-intensive offloading refers to sending heavy and complex computational tasks to remote cloud servers or edge servers while keeping data processing closer to end-user devices. This technique is ideal for AR/VR applications that require real-time rendering and sophisticated algorithms for object detection and scene generation [123].

Offloading computation-intensive [124] tasks to powerful servers can enhance graphics rendering and maintain high frame rates. However, it may also introduce latency due to communication delays. In contrast, data-intensive offloading [125] emphasizes transmitting large volumes of data, such as 3D models and sensor data, to the edge or remote servers for processing. This technique is suitable when the AR/VR device has limited processing power but has a reliable high-bandwidth connection [124]. It could waste network resources and introduce latency concerns.

Similarly, computation-intensive caching involves storing computationally expensive tasks or their results on edge nodes or local devices. By keeping these computations close, subsequent interactions can be executed more efficiently. This technique signifi-

cantly reduces the need for resource-intensive calculations, leading to notable performance improvements. This could be impractical (unfeasible) to perform edge devices with limited resources [32]. On the other hand, data-intensive caching [126] is related to storing substantial volumes of data on edge devices or servers. These data may encompass high-resolution textures, 3D models or other vital information for AR/VR applications [127]. Researchers mention that a balanced approach that combines computation-intensive and data-intensive caching may be the most effective. This hybrid technique optimizes local device resources while leveraging edge and cloud servers when necessary. The choice between these techniques should align with the specific requirements and constraints of the AR/VR application and the overall system architecture.

5.3.2. Binary vs. Partial Offloading/Caching

Binary offloading transfers an AR/VR application or workload entirely to the cloud/edge or runs it entirely on the local device [128]. Binary offloading [128] is suitable for applications where a clear separation between local and remote processing is feasible. For example, some tasks may require cloud-based processing, while others can be handled locally. However, it may not be optimal for applications with dynamic resource requirements, as transferring an entire application can introduce delays. On the contrary, partial offloading involves dividing an AR/VR task into subtasks or components, and then offloading these subtasks to edge or cloud servers [129]. Partial offloading is ideal for applications with heterogeneous resource needs, where some subtasks can be executed more efficiently on the local device, while others could benefit from cloud or edge processing. It optimizes resource utilization by offloading only subtasks that benefit from cloud/edge processing, and minimizes latency by keeping latency-sensitive subtasks local. It also allows applications to adapt to changing conditions and resource availability. However, partial offloading [130] requires complex analysis and decision-making algorithms to determine which subtasks to offload, where to execute them, and how to manage their dependencies.

Similarly, binary caching involves caching the entirety of the applications or datasets in an all-or-nothing manner. It may not be practical for applications with large datasets that exceed the storage capacity of edge devices, and its bandwidth requirements for initially caching the entire dataset can be high. Partial caching involves selecting specific portions of applications or datasets based on predefined criteria. Partial caching [131] is more practical; however, it requires more sophisticated cache management algorithms to determine which data to cache, when to evict cached items, and to handle dependencies among cached elements.

5.4. Service Migration

Service migration refers to moving services from one location (e.g., edge server) to another (e.g., cloud server) [24]. Service migration within AR/VR systems must engage with complex issues specific to AR/VR itself. In fact, maintaining a consistent flow in this migration process is crucial for AR/VR systems. This means that when a service is moved or relocated, it should be done in a way that does not disrupt the AR/VR user experience [132]. Another challenge appears when end devices (e.g., smartphones) are mobile because it adds another layer of complexity to service migration. As users change their physical location, the application may need to migrate services to different servers or data centers to maintain optimal performance [133]. Therefore, service migration must adapt to dynamic changes without degrading user experience.

5.4.1. Preemptive vs. Reactive Migration

Preemptive migration involves the planned migration of services before performance issues or resource constraints occur [134]. This type of migration is usually based on

predictive analysis or policy-based decisions [135]. Preemptive migration can be used to efficiently balance workloads across servers or prevent overloading specific nodes. It is particularly valuable for systems that seek to maintain stable and efficient operations by anticipating future requirements and potential challenges.

Reactive migration occurs when services encounter performance degradation, hardware failures, or other unexpected circumstances [136]. Reactive migration aims to recover from faults, performance bottlenecks, or degraded qualities by migrating services to more suitable resources. Reactive migration is performed using predictive and prescriptive analytics to ensure service availability and quality in unexpected and damaging conditions.

5.4.2. Live vs. Non-Live Migration

Live migration is applied for moving services while they are currently serving (actively running) users or applications [137]. This technique is characterized by minimal interruptions or downtime, making it ideal for AR/VR systems. Live migration is commonly used to optimize resource usage, and cloud service relocation in response to changing demands without affecting user access. In contrast, non-live migration requires stopping the services before starting the migration process [138]. During the downtime, users may experience service unavailability and interruptions. To minimize risks associated with data corruption or errors, non-live migration is often used to migrate large databases, to perform major software updates, and/or to replace hardware components. The choice between live and non-live migration depends on factors such as the need for continuous service availability, data integrity requirements, migration complexity, and tolerance for service downtime.

5.5. Security and Privacy

Edge–cloud AR/VR systems bring many security and privacy challenges that must be addressed comprehensively. One of the primary issues is protecting the user data, because these systems often involve transmitting and storing sensitive data over the edge–cloud network, where unauthorized access to personal and private data could be possible. Malware, cyberattacks, and vulnerabilities in edge AR/VR devices can also threaten the security, privacy, or even functionality of these systems. Identifying and mitigating these threats are critical for providing the stability and safety of edge–cloud AR/VR applications. For example, consider a situation where the system experiences sensory overload attacks, where the overwhelming sensory AR/VR inputs can disrupt the system’s regular operation [44].

The issue of user privacy is also a large threat. Privacy violations and data abuse concerns become more noticeable as AR/VR systems collect and process user data, including location, biometrics, and behavioral patterns [44]. Finding a balance between providing high-quality services to AR/VR users and respecting their privacy is a complex challenge and requires careful considerations. It is important to note that data privacy regulations continually evolve, adding complexity to the design of AR/VR systems. Compatibility with data protection regulations, such as the General Data Protection Regulation (GDPR) [139] in Europe or the California Consumer Privacy Act (CCPA) [140] in the United States, are essential tools to ensure fulfilling privacy concerns. Failure to meet these legal requirements can have significant legal and financial consequences [7]. Security/privacy techniques can be applied on three levels: input, data, and output.

5.5.1. End-to-End Security and Privacy in Edge–Cloud AR/VR Systems

Input-level security/privacy is a critical aspect that revolves around protecting the data obtained from the sensors on the AR/VR device. For example, the device’s camera might capture personal videos, containing private and confidential content. These sensitive data must remain inaccessible to unauthorized parties. Therefore, input-level techniques

are implemented to ensure that sensitive data are protected, that is, neither exposed nor shared with unwanted entities or unauthorized parties.

In most cases, collected data from different sensors are stored within a database and processed to generate outputs. Data level security/privacy techniques (also known as data access security or data protection) address concerns throughout this data lifecycle, including data collection, aggregation, processing, and storage [141]. Within AR/VR systems, one of the primary concerns is to prevent unauthorized applications from accessing stored data (malicious or unintentional). Data-level security/privacy techniques help maintain the trustworthiness of the AR/VR system and safeguard sensitive information from misuse or violations.

The potential threat that malicious parties or applications pose sometimes relies on their ability to inject or manipulate the output stream of a system [7]. When such unauthorized access occurs, it opens the door to modifying the output stream. For example, consider an AR/VR application designed for remote repair or maintenance. Malicious interference might cause the application to provide incorrect instructions or deliver the correct instructions in the wrong sequence, ultimately leading to a failure in the system. Therefore, a robust output-level security and privacy technique prevents malicious entities from trying to manipulate the output stream.

5.5.2. Existing Research

Several works focus specifically on the security and privacy of edge–cloud systems. For AR/VR-based use cases, Braeken et al. [142] designed an architecture known as ES-SMAR, providing a solid foundation to protect the confidentiality of AR/VR data in the healthcare sector. To address security and privacy challenges, they presented a secure key management scheme that includes the registration and authentication phases. Qiao et al. [143] introduced a partitioned histogram data publishing algorithm that uses the wavelet transform to protect the privacy of edge users. The approach involves using a greedy algorithm for partitioning, followed by applying a wavelet transform for introducing noise. The final step focuses on preserving the histogram’s authenticity and usability by restoring the original histogram’s structure through reduction. Sharma et al. [144] also studied a data protection approach that employed a fuzzy-based convolution neural network with the Laplace mechanism to inject noise into the data in edge–cloud environments. Guzman et al. [44] provided a comprehensive survey on security and privacy aspects and solutions in AR/VR systems. Furthermore, some survey papers (e.g., [145–147]) focus on the privacy and security aspects of edge–cloud computing from more generic (less focused on AR/VR) points of view.

6. Solutions

In this section, we provide an overview of various solutions introduced for technical aspects related to service placement, task offloading/caching, and service migration. We chose not to conduct a detailed investigation into solutions for other technical aspects, such as communication technologies, security, and privacy, as these require further exploration and are beyond the scope of this paper.

Based on our review of the literature, solutions for service placement, task offloading/caching, and service migration can be categorized into three main types: heuristics, metaheuristics, and ML-based approaches. Some papers also propose hybrid solutions that combine these methods.

6.1. Heuristic-Based Solutions

Heuristic methods [123,129,148–151] are a category of problem-solving techniques that aim to produce good enough solutions within a reasonable runtime, especially when finding an exact solution is impractical because of time or computational constraints. Heuristics play an important role in finding solutions where fast and efficient decision-making is required and the system is highly dynamic. Although heuristics are computationally efficient, making them well-suited for real-time decisions in edge computing, they do not guarantee the optimal solution. Many heuristic methods focus on local optimization without considering the global impact of their decisions.

6.2. Metaheuristic-Based Solutions

Metaheuristics are strategies [148,152–154] designed to explore the solution space of complex optimization problems more effectively than traditional heuristics. Unlike simple heuristics that may get stuck in local optimal points, metaheuristics are designed to explore the global solution space, increasing the probability of finding a near-optimal or optimal solution. Although metaheuristics are designed to find good solutions efficiently, they can still be computationally expensive, especially for very large or complex problems that might limit their applicability in real-time edge computing scenarios where quick decisions are crucial. In addition, many metaheuristics require careful tuning of parameters (e.g., mutation rates in genetic algorithms, and cooling schedules in simulated annealing), which can be challenging and may affect the performance of the algorithm.

6.3. ML-Based Solutions

ML-based solutions [155–159] represent a data-driven approach that leverages the power of ML models to predict, optimize, and make real-time decisions regarding service placement, task offloading/caching and service migration in edge computing environments. The ML-based approaches can be based on supervised learning, unsupervised learning, or Reinforcement Learning (RL). In supervised learning, models are trained on labeled datasets, where the input data are paired with the proper output. The model learns to map inputs to outputs and can then predict the output for new and unseen data. Unsupervised learning models are used to find hidden patterns or structures in unlabeled data. These models do not require labeled output, making them useful for clustering, anomaly detection, and data reduction. RL involves training an agent to make decisions by interacting with an environment and receiving feedback in the form of rewards or penalties. The agent learns to maximize cumulative rewards by optimizing its actions over time.

All three types of machine learning approaches can be applied to address the technical challenges of edge–cloud computing. For example, supervised learning can be applied to predict the workload of an edge node based on historical data. A model could be trained to forecast the CPU or memory usage of a node by considering factors such as the type of tasks or network traffic. This enables the system to make proper decisions, ensuring that nodes are provisioned with the appropriate amount of resources to handle the anticipated load. Clustering algorithms, as an example of unsupervised approaches, can group similar tasks or edge nodes based on their characteristics. Tasks with similar resource requirements can be clustered together, facilitating more efficient batch processing or resource allocation. RL can also be employed to dynamically manage resources in an edge network. For example, an RL agent could learn to allocate services to different edge nodes in real-time, based on factors such as current load, network latency, and energy consumption. The agent would receive rewards for maintaining low latency and high throughput while minimizing energy usage. Over time, the agent would learn optimal strategies for service placement, even in the face of unpredictable changes in workload or network conditions.

6.4. Hybrid Approaches

Hybrid approaches [160–163] combine two or more solution paradigms (such as heuristic, metaheuristic, and ML-based methods) to leverage their complementary strengths in edge–cloud optimization. Common hybrid designs use learning-based models to guide or initialize heuristic or metaheuristic searches, or apply metaheuristic optimization to tune model parameters or decision policies. These approaches aim to achieve a balance between solution quality, adaptability to dynamic environments, and computational efficiency. Although hybrid methods demonstrate improved performance compared to single-paradigm solutions, they may also introduce additional system complexity and design overhead, particularly in large-scale or real-time edge–cloud scenarios.

Table 2 shows the classification of the reviewed references according to system architecture, communication technologies, technical aspects, and solution approaches.

Table 2. Classification of the reviewed references according to system architecture, communication technology, technical aspects, and solution approaches.

Criterion	Category	References
Architecture	Cloud-based	[45–50]
	Edge-based	[52–64]
	Edge–cloud-based	[67–80]
Communication technology	Wired	[6,63,97]
	Wireless	[82,90–96]
	Hybrid	[49,51,52,55]
Technical aspects	Task offloading	[32,120,124,125,128–130]
	Caching	[121,122,126,127,131]
	Service placement	[23,99–112,114–116,118]
	Migration	[24,134–138]
	Security/Privacy	[44,142–147]
Solution approaches	Heuristic methods	[123,129,148–151]
	Metaheuristic methods	[148,152–154]
	ML-based methods	[155–159]
	Hybrid methods	[160–163]

6.5. Discussion and Analysis of the Solutions

Although heuristic, metaheuristic, and learning-based approaches have been widely adopted for service placement, task offloading, and service migration in edge–cloud AR/VR systems, their effectiveness strongly depends on the trade-off between decision latency and optimization quality. Heuristic algorithms (e.g., greedy, rule-based, or priority-driven methods) show low computational complexity and fast decision-making, making them well suited for latency-critical AR/VR workloads in which decisions must be made within milliseconds to satisfy strict delay constraints. However, their limited search space and reliance on handcrafted rules often result in sub-optimal resource utilization under highly heterogeneous edge–cloud conditions. Metaheuristic algorithms, on the other hand, explore a broader solution space and typically achieve higher-quality placement or offloading

decisions. Nevertheless, their iterative nature introduces non-negligible computational overhead and convergence delay, which can conflict with the real-time responsiveness required by AR/VR applications. As a result, these techniques are more suitable for offline planning, periodic re-optimization, or relatively static environments rather than per-request online decision-making.

Learning-based approaches, particularly reinforcement learning and lightweight neural models, attempt to balance these trade-offs by shifting most of the computational cost to a training phase and enabling fast inference at runtime. Although such approaches can achieve near-heuristic latency during inference, their effectiveness depends on the stability of the environment and their ability to adapt to workload drift, which remains a challenge in highly dynamic AR/VR scenarios. In addition, as discussed above, hybrid approaches combine complementary optimization techniques to balance latency sensitivity and solution quality, which is a critical requirement in AR/VR systems. Hybridization is particularly effective in AR/VR workloads because real-time interactions demand immediate responses, whereas infrastructure-level optimization can tolerate higher computational overhead. For instance, a heuristic approach may select the nearest edge server to satisfy strict latency constraints, while a genetic or learning-based optimizer periodically adapts placement strategies to evolving workload distributions. Despite their advantages, hybrid methods introduce additional system complexity and coordination overhead, requiring careful synchronization between fast and slow decision layers. Nevertheless, they represent a promising compromise for AR/VR systems in which neither purely heuristic nor purely optimization-driven solutions are sufficient.

7. Analysis of Recent Research Trends on the Technical Aspects

This section goes beyond a quantitative literature analysis by synthesizing recent research trends across the identified technical aspects of edge–cloud AR/VR systems. The analysis highlights technology maturity levels, emerging solution directions, and underexplored challenges, thereby providing a roadmap-oriented perspective for guiding future research and system design.

Based on our second search methodology (discussed in Section 3), in this section, we conduct an additional review of the literature with the aim of investigating recent research trends in the field and their relation to each technical principle discussed above. In fact, in this section, we aim to analyze the research trends in each technical aspect and investigate the number of papers related to these aspects. Our goal is to provide a comprehensive overview of how different technical elements have been explored in recent studies. Specifically, we examine which network architectures, performance metrics, objectives, communication models, implementation types, and methods have been used in the reviewed papers. This analysis will help to determine whether certain approaches, architectures, or evaluation metrics are dominant and whether there is a need for further exploration of under-represented areas.

The analysis presented in this section is based on studies published within the last five years, providing a focused view of recent research directions and emerging topics in edge–cloud systems for AR/VR workloads. According to the second search methodology, we found that 73 articles were explicitly related to the technical aspects of AR/VR, whereas 106 articles generally focused on the technical aspects of edge–cloud computing. Out of the reviewed papers, only 12 focused on a specific use case. It is worth mentioning that some works cover more than one technical aspect (e.g., [65]).

A substantial number of studies focus on the topic of offloading in AR or VR systems. In contrast, research on other important aspects, such as service migration, security, and privacy, remains significantly less prevalent. Relatively few studies have addressed

the challenges associated with real-time service migration. Similarly, security and privacy considerations, despite being critical in AR/VR systems, have not received as much attention. The findings highlight a research gap where offloading has been extensively studied, but other crucial aspects such as service migration, security, and privacy require further attention to ensure robust, secure, and adaptive AR/VR experiences in edge–cloud computing. (see Figures 6 and 7). The observed trends indicate that while certain technical aspects (such as task offloading) have reached a relatively mature stage, others, including service migration and security/privacy mechanisms, remain underexplored. These findings provide clear guidance on where future research efforts should be prioritized, effectively serving as a technology roadmap for the evolution of edge–cloud AR/VR systems.

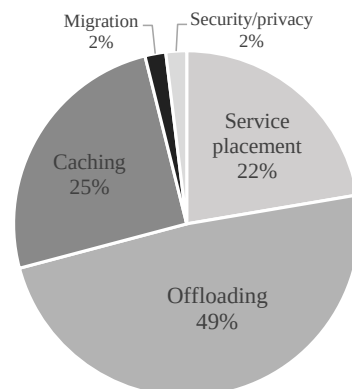


Figure 6. The proportion of recent AR/VR-specific works on the technical aspects based on the second search methodology.

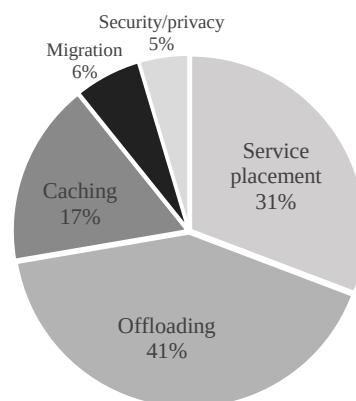


Figure 7. The proportion of recent generic works on the technical aspects based on the second search methodology.

Figure 8 provides further insight, showing that a significant number of research works take a broad perspective on the problem, with only a smaller proportion focused on specific AR/VR use cases. Recent research trends also favor the investigation of edge–cloud architecture driven by advancements in communication technologies. In particular, 5G technology is popular in most recent work. It is important to mention that concepts are interchangeably used in some works. For example, offloading and service placement are considered the same concept, and some studies examine multiple aspects simultaneously. It should also be noted that some papers serve a specific purpose, such as delivering services with ultra-low latency, optimizing energy consumption, balancing the network load, and increasing operational throughput. The primary objectives of most related studies were to reduce delay and provide services with ultra-low latency.

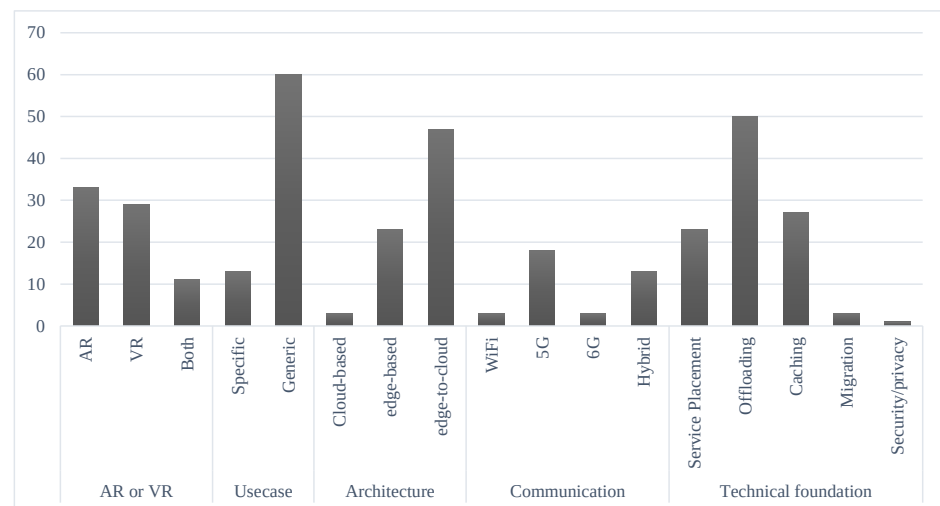


Figure 8. Analysis of the recent papers regarding technical aspects based on the second search methodology.

In addition, as illustrated in Figure 9, the majority of the reviewed studies focus on achieving a single primary objective, with latency minimization being the most commonly addressed concern. Given the strict real-time processing requirements of AR/VR applications, reducing latency is a critical challenge, as even minor delays can significantly degrade user experience. Consequently, a large portion of the literature prioritizes optimizing system latency. However, a subset of studies adopts a multi-objective approach, considering multiple performance metrics simultaneously. Among these, the most frequently addressed combination involves both latency and energy consumption. This trend reflects the growing recognition that while reducing latency is essential, it must often be balanced with energy efficiency, particularly in edge devices with limited power resources. Some works also incorporate additional objectives such as system reliability, computational load balancing, and cost-effectiveness, though these considerations are less prevalent.

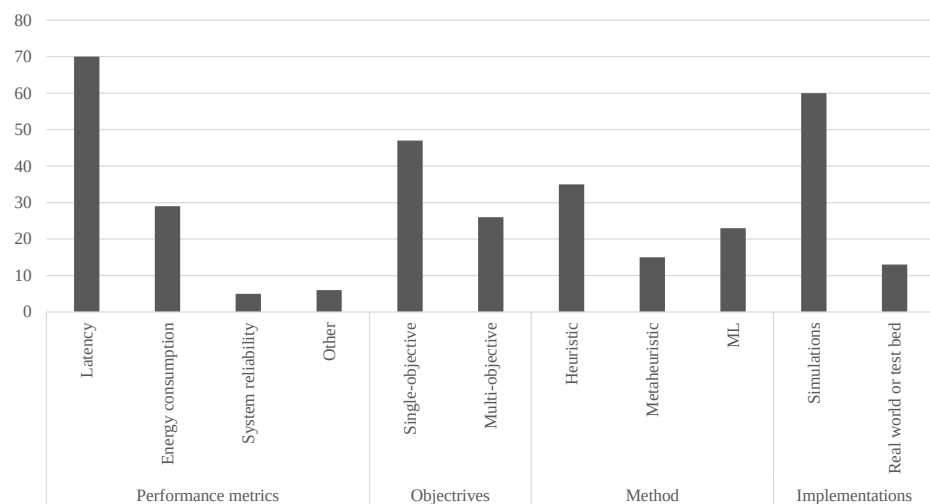


Figure 9. Analysis of recent papers based on performance metrics, objectives, methods used, and implementation types.

Another notable observation is that in AR/VR-specific papers, heuristic methods are more commonly employed to address the challenges of edge–cloud AR/VR systems. AR/VR-focused research tends to use heuristic approaches because of their ability to provide reasonable solutions with significantly reduced computational overhead. Given

the real-time constraints of AR/VR applications, heuristics are often preferred as they enable rapid decision-making while maintaining acceptable levels of performance. In addition to heuristics, machine learning-based approaches are also increasingly utilized in AR/VR-specific studies. The growing adoption of machine learning reflects its potential to optimize performance in highly dynamic edge–cloud environments. However, meta-heuristic and exact optimization methods are less frequently employed in AR/VR-specific research. The primary reason for this is the high computational runtime associated with these techniques (despite near-optimal solutions provided by metaheuristics), which can negatively impact AR/VR applications in terms of both latency and energy consumption.

Moreover, the majority of studies relied on simulations to evaluate their approaches. A small portion of the reviewed papers conducted real-world experiments to validate their findings. This indicates that while simulation-based evaluations are widely adopted due to their scalability and ease of implementation, there remains a gap in real-world testing, which is essential for assessing the practical feasibility and deployment challenges of AR/VR applications in edge–cloud environments.

Figure 10 illustrates the growth of research activity across core technical aspects of edge–cloud systems over the period 2020–2025. The results show a pronounced growth in studies on task offloading and service placement, indicating their central role in addressing latency and computational constraints. Research on caching shows a steady increase. In contrast, service migration and privacy/security remain comparatively less explored, despite their importance in dynamic and data-intensive edge–cloud environments. These trends highlight both mature research areas and underexplored topics, providing a quantitative basis for identifying future research opportunities.

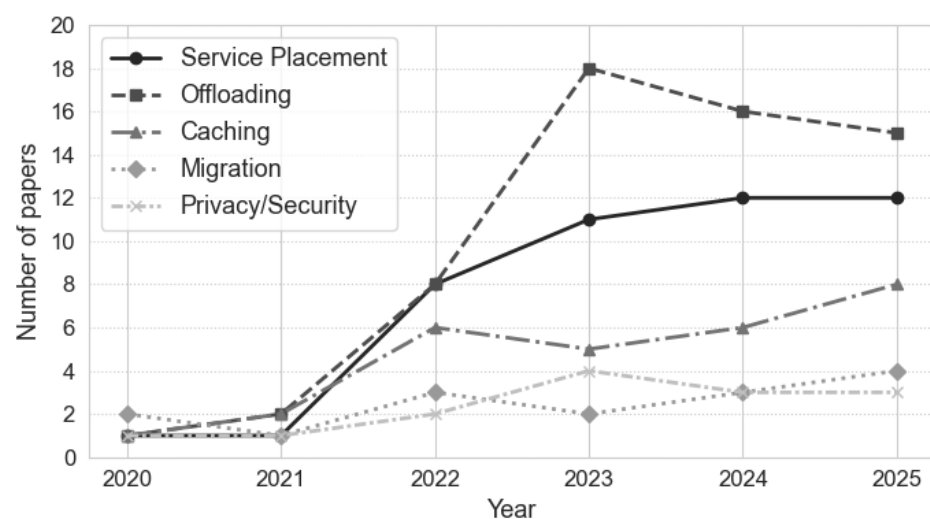


Figure 10. Publication trends across core technical aspects of edge–cloud systems for both AR/VR-specific and general use cases.

It is also worth noting that the majority of the selected studies were retrieved from IEEE Xplore (84 papers), followed by ScienceDirect (53 papers) and the ACM Digital Library (42 papers), reflecting the dominant databases for system-level research on edge–cloud AR/VR technologies.

8. Open Research Directions

Ultra-low latency and high-bandwidth communication: Providing real-time data transmission for AR/VR applications remains a fundamental open research challenge. Although current wired and wireless communication technologies can meet ultra-low

latency requirements, they struggle to simultaneously support the extremely high data rates demanded by immersive AR/VR workloads. Addressing this challenge requires innovative network designs, including novel communication protocols, advanced video coding techniques, and the integration of specialized hardware and software capabilities at the edge layer. Solving these issues is critical to ensuring immersive user experiences while maintaining high data throughput.

Online and adaptive service placement: Despite the growing body of research on service placement in edge–cloud environments, developing cohesive and consistent placement strategies for real-world AR/VR systems remains challenging. Practical AR/VR use cases are highly dynamic, involving mobile users, fluctuating workloads, and heterogeneous resources. Consequently, placement solutions must operate in an online manner, rapidly adapt to changing system states, and seamlessly integrate both mobile and non-mobile devices. Designing efficient, agile, and scalable online placement algorithms for edge–cloud AR/VR systems therefore represents a key open research direction.

Task offloading, caching, and service migration: While offloading and caching have been extensively studied in general edge computing contexts, their specific implications for AR/VR applications remain underexplored. Emerging communication technologies (e.g., 5G and beyond) introduce new architectural possibilities and resource distributions that must be explicitly considered when designing offloading and caching strategies for AR/VR workloads. In addition, existing service migration approaches largely focus on generic and static scenarios, often relying on preemptive or non-live migration techniques. There is a notable lack of research on reactive and live service migration tailored to the stringent latency and continuity requirements of AR/VR applications in dynamic edge–cloud environments.

Security and privacy in AR/VR edge–cloud systems: Security and privacy remain critical but under-addressed challenges in AR/VR edge–cloud systems. In particular, protecting user-generated content (such as audio, video, sensor readings, and location data) requires robust encryption and secure communication protocols across devices, edge nodes, and cloud infrastructures. Furthermore, reliable authentication and identity management mechanisms are essential to prevent unauthorized access and ensure user trust. Addressing these concerns opens important research directions at the intersection of edge computing, AR/VR, and cybersecurity.

AI-driven adaptive management: Artificial intelligence, and machine learning in particular, offers significant potential for addressing the aforementioned challenges in a unified manner. An important future research direction lies in developing AI-based approaches capable of dynamically and autonomously determining optimal policies for service placement, task offloading and caching, and service migration in real time. Such approaches must balance decision quality, computational efficiency, and robustness to changing conditions in order to meet the stringent performance requirements of AR/VR applications.

9. Conclusions

Edge–cloud computing offers many possibilities but also presents numerous challenges. This paper provided a structured technical analysis of edge–cloud systems for AR/VR workloads, focusing on the core architectural and system-level aspects driven by their computationally intensive and ultra-low-latency requirements. By classifying and examining communication technologies, service placement, task offloading and caching, service migration, and security and privacy, this study highlighted how AR/VR constraints reshape design trade-offs in edge–cloud environments. The analysis of existing solutions and recent research trends revealed both mature areas and underexplored challenges, particularly in migration and security under real-time constraints. The insights and research

directions presented in this work aim to guide future efforts toward more adaptive, scalable, and reliable edge–cloud systems capable of supporting demanding AR/VR workloads.

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